



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

EE6405

Natural Language Processing

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Deep-dive into NLP Applications



Overview

- **We will take a closer look at how NLP is used in large ML projects**
- **Two NLP-base applications**
 - Fraud Detection Model
 - Human Trafficking Model
- **Two ChatGPT-based applications (with prompt engineering):**
 - Content Moderation
 - Product Recommendation Chatbot

Fraud Detection Model

Scenario

- On e-commerce platforms, there are often fraudulent sellers.
- They typically commit two types of fraud, both can be detected using textual data

Fraud Type	Data Source
Fulfillment Fraud	Product review from buyer
Off-platform Fraud	Messages from seller to buyer

Goal

- Use NLP combined with other ML techniques to build fraud models.

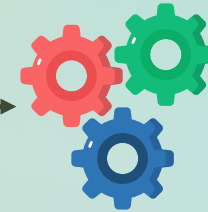
Fulfilment Fraud Model

"Absolutely furious! The product I received is a complete fake! It's nothing like what was advertised on the website. The quality is horrendous, and it's clear that this is a cheap knockoff. I feel completely cheated and misled by the seller. Buyers, beware – do not trust this seller! They are selling counterfeit items and taking your money. I've reported this to the platform and I expect a full refund!"

"This is a warning to all potential buyers: DO NOT BUY from this seller! The product I received is an absolute sham – it's fake and of incredibly low quality. The seller is a fraud, selling counterfeit products at the price of the real thing. I've been completely ripped off and I'm so angry that I fell for this scam. I demand a refund and will be taking this matter further. This seller should be banned from the platform for their dishonest and deceptive business practices!"

Product Reviews highlighting Fulfilment Fraud

**Aggregated
at seller level**



**Sentence
Transformer
Model**

**Sentence
Embeddings**

**Each sentence gets
a high dimensional
embedding**

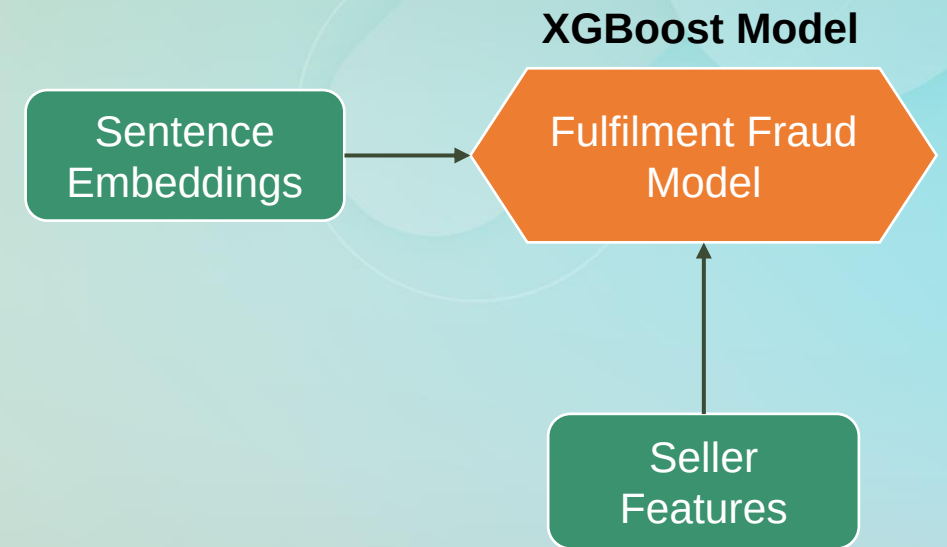
**Fulfilment Fraud
Model**

XGBoost Model

Fulfilment Fraud Model

Technical Details

- There are many freely available Sentence Transformers, a popular one is provided by the Hugging Face package.
- We can customize the embedding size, popular choices are 512 and 768 and 1024.
- For supervised learning, XGBoost is well-suited for dealing with embedding data. If the data is sparse, XGBoost can also provide an efficiency gain.
- We can further improve our model by incorporating other seller features.



Fulfilment Fraud Model



Downstream Process

- Suspected fraud sellers are sent for review by human investigators.
- Two levels of reviewers are used to ensure correctness.
- Confirmed fraud sellers are punished. Different level of punishments are available depending on severity.

Human Trafficking Model

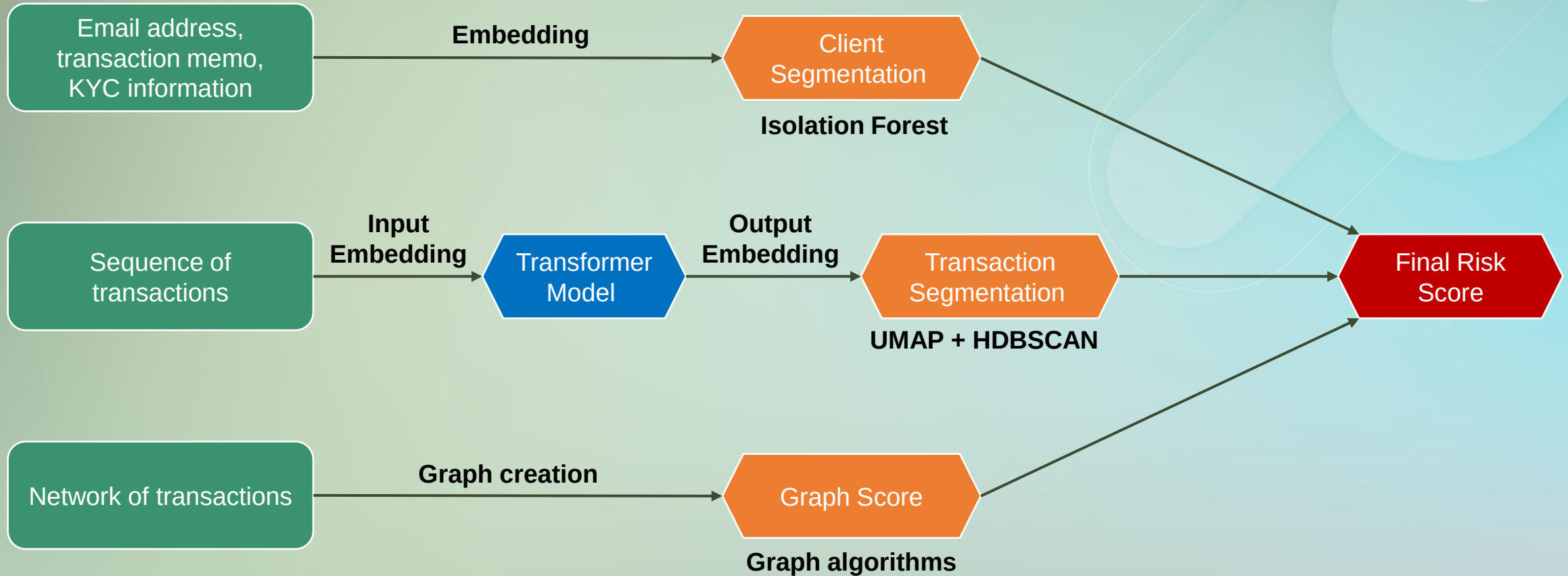
Scenario

- Human trafficking is a dark business that is worth an estimated \$150 million USD annually.
- Some of these dirty money is laundered through major banks.
- Leveraging AI/ML on bank's data, we can detect human trafficking activities.

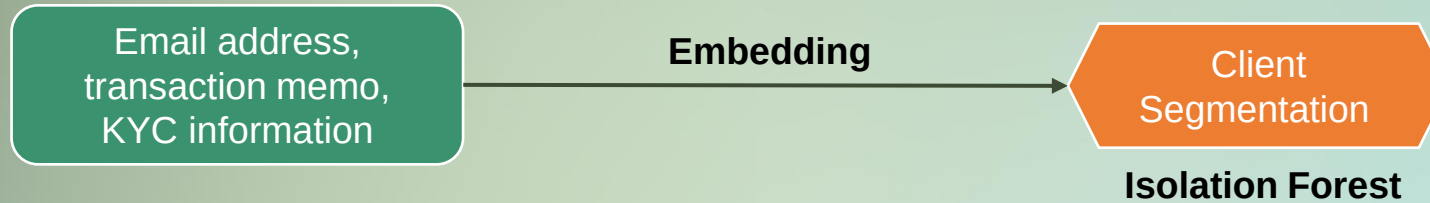
Goal

- Use NLP combined with other ML techniques to build a human trafficking detection model.

Human Trafficking Model

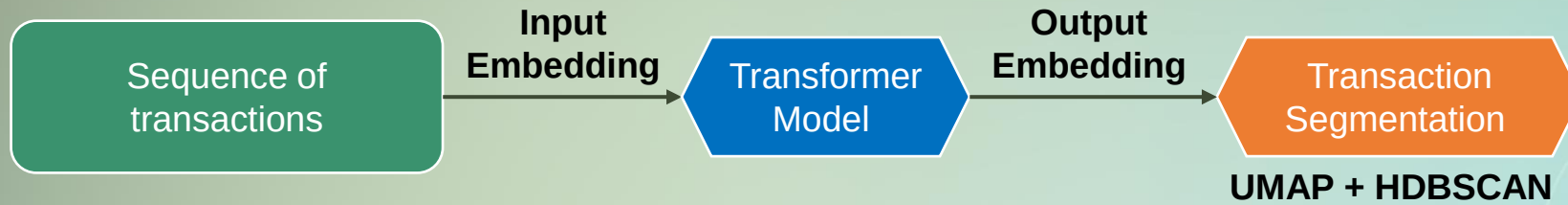


Human Trafficking Model



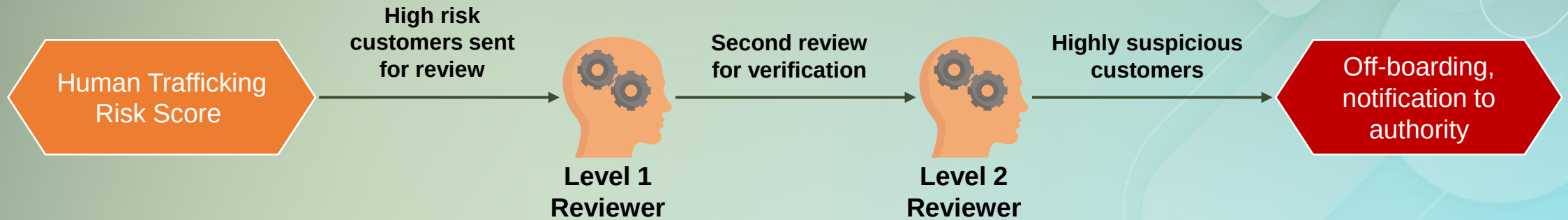
- This human trafficking model is specifically looking for cases of sex trafficking.
- In the country where model was built, it is common to transfer money via a bank by providing an email address. This is called Email Money Transfer (EMT).
- An additional transaction memo is added as part of the transfer.
- KYC information, such as occupation of the sender and receiver are also parsed.
- All these text information was combined into an embedding.
- Pre-trained BERT was used produced the embedding. This uses 768 dimensions.
- Isolation Forest technique is used to cluster the embeddings.

Human Trafficking Model



- We learned that the transformer framework is used to process a sequence of text
- Here instead of sequences of text, we have sequences of transactions.
- The output of a transformer model is a transformed embedding of the input sequence.
- These transformed transactions are then clustered to produce transaction segments.
- Uniform Manifold Approximation and Projection (UMAP) and Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) are used to perform the clustering.

Human Trafficking Model



Downstream Process

- High risk customers suspected of potential human trafficking activities are sent for review by human investigators.
- The two levels of investigators are from the bank's special investigations unit, specifically handling challenging financial cases.
- Highly suspicious customers can be off-boarded, with extreme cases notified to relevant authorities.

Content Moderation

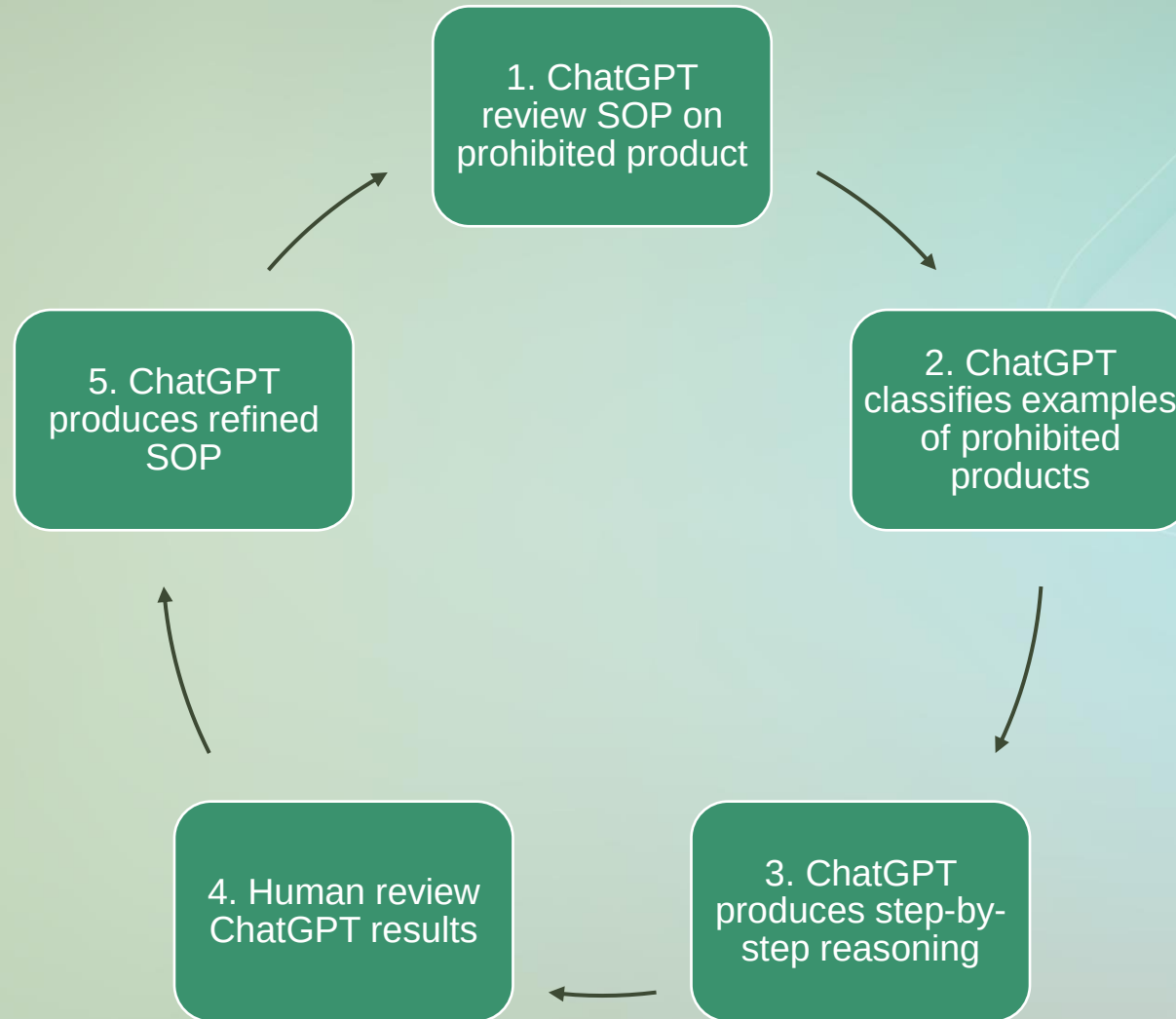
Scenario

- On social media and e-commerce platforms, there are regulatory needs to moderate content.
- On social media, various images and videos are prohibited.
- On e-commerce platforms, certain items are prohibited from listing.

Goal

- Use the power of ChatGPT and prompt engineering to effectively screen for prohibited items.

Prohibited Product Example



Sequence of Prompts

- **Provide input to define objective**

"Read this Standard Operating Procedure (SOP) document which defines 'Alcohol and Tobacco'. [Insert the text of the SOP here, or specify how you're providing it, e.g., through a document upload or a link]. Please summarize the key characteristics that the SOP uses to define 'Alcohol and Tobacco'."

- **Vital here to ask ChatGPT to summarize the key characteristics from the document**

Sequence of Prompts

- **Provide case and request for analysis**

"Here is a product description: [Insert product description]. Based on the SOP definition of 'Alcohol and Tobacco' you summarized, does this product fall under the category of 'Alcohol and Tobacco'? Answer with 'Yes', 'No', or 'Unclear'."

- **Important to provide ChatGPT with a catch-all of “Unclear”**

Sequence of Prompts

- **Request step-by-step reasoning**

"Provide a step-by-step reasoning for your previous answer about the product classification, detailing how you used the SOP definition to determine if the product is 'Alcohol and Tobacco'."

- **Important to ask ChatGPT to refer back to the SOP**

Sequence of Prompts

- Request feedback and revision

"Based on a series of product classifications (Yes, No, Unclear) that you previously provided and which have now been reviewed and corrected by a human expert, we need to update the SOP definitions to better align with these reviewed answers. The goal is to ensure that future classifications of products as 'Alcohol and Tobacco' can be logically reasoned from the updated SOP. Please revise the SOP to reflect the insights gained from the corrected classifications, focusing on clarifying any ambiguities and ensuring that the criteria are precise and align with the reasoning used in the reviewed answers."

Prompt Engineering Keys

Clarity of Objective:

- Define a clear, specific goal for each prompt. This ensures the AI understands exactly what is expected and can generate relevant responses.

Sequential Logic:

- Arrange prompts in a logical sequence that guides the AI through a thought process or a set of tasks.

Specificity and Detail:

- Include detailed information and specific instructions in prompts to minimize ambiguity and guide the AI's response accurately.

Feedback Incorporation:

- Use the AI's responses to refine and adjust subsequent prompts, ensuring better accuracy and relevance in the answers.

Recommendation Chatbot

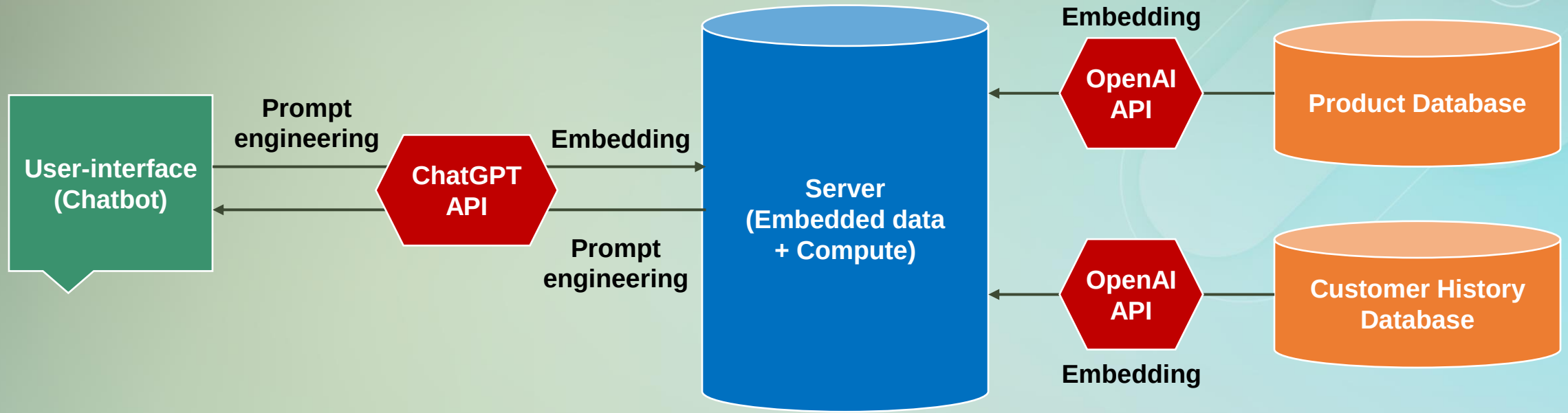
Scenario

- Before ChatGPT, chatbots only have fixed sets of responses.
- With ChatGPT, chatbots can be used to guide users to a large range of desired product and services.

Goal

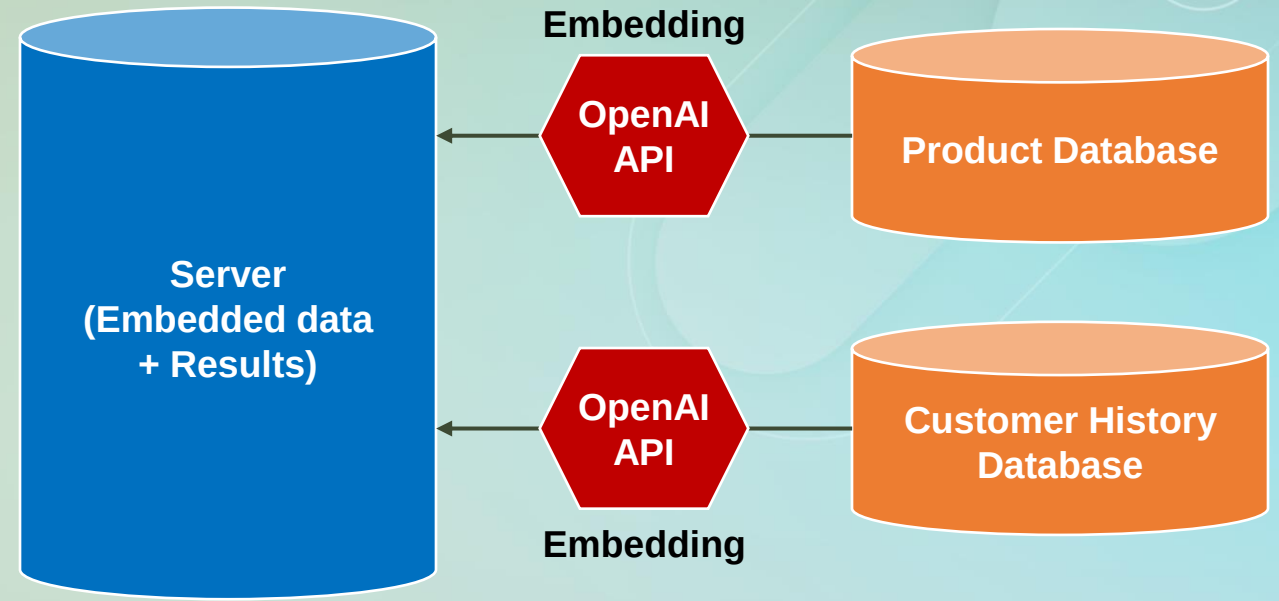
- Use the power of ChatGPT's API and prompt engineering to create an intelligent chatbot for product recommendation.

Recommendation Chatbot



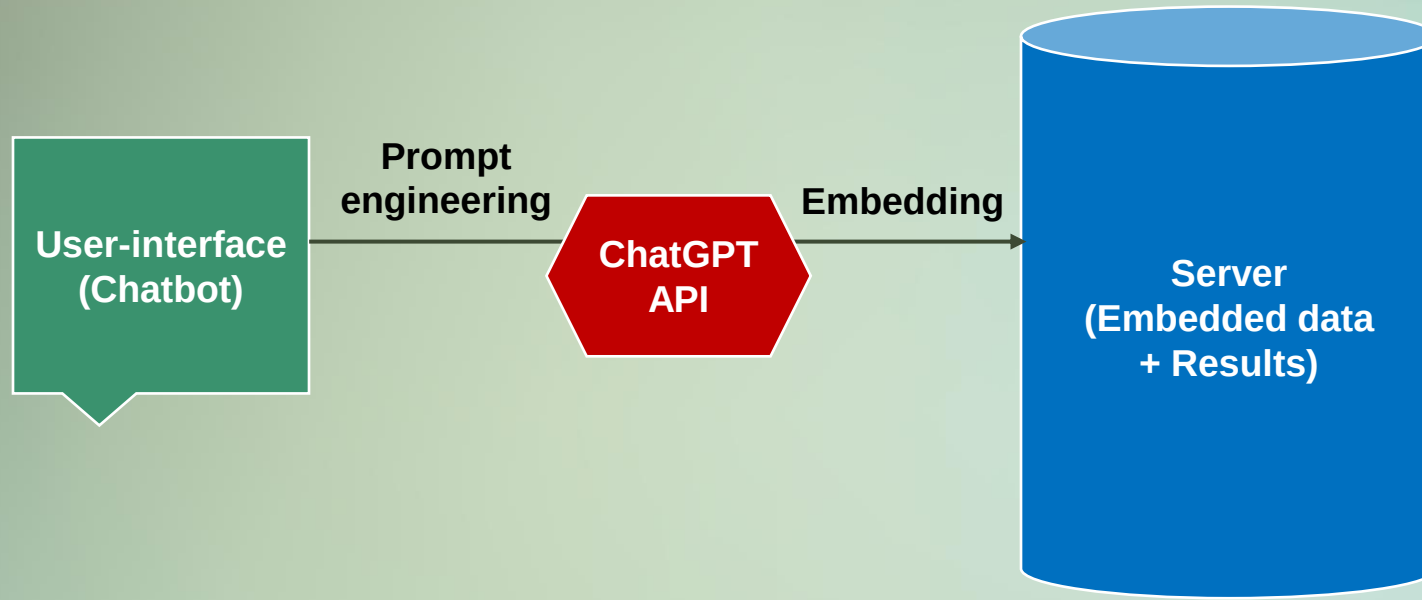
Recommendation Chatbot

- For each product information or customer history, an embedding can be created by calling OpenAI's API
- The set of embeddings are saved for future use



```
# Generate embeddings using the OpenAI API
product_data_df['text_embedding'] = product_data_df.combined.apply(lambda x: get_embedding(x, engine='text-embedding-ada-002'))
```

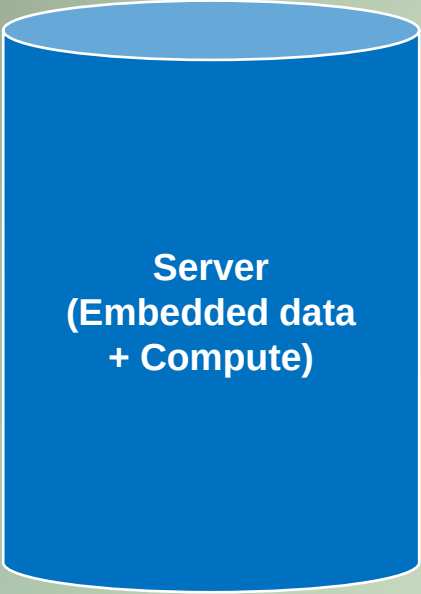
Recommendation Chatbot



- Customer interact with Chatbot powered by ChatGPT.
- The relevant information (e.g. product request) from the customer is extracted
- An embedding for the customer input is created

```
# Generate embeddings for the customer input message
response = openai.Embedding.create(
    input=customer_input,
    model="text-embedding-ada-002"
)
embeddings_customer_question = response['data'][0]['embedding']
```

Recommendation Chatbot



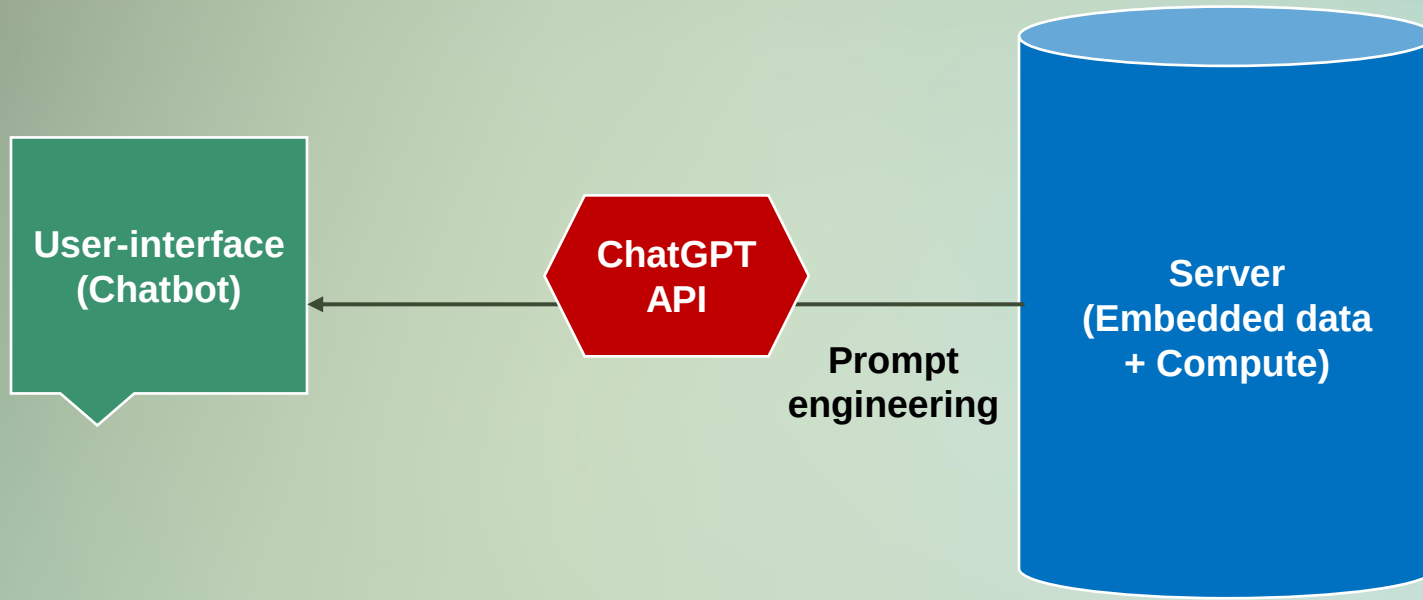
Server
(Embedded data
+ Compute)

- With the embedding of the customer input, we can first find if the customer ever purchased the same or similar product.
- We can do this with a similarity measure. Cosine similarity, which is a very useful similarity for high dimension data, is often used.
- If the customer did not make a similar purchase, we can search for all available products and find the best recommendation.

```
# Calculate similarities between customer's previous purchases and the customer's chat input question
customer_order_df['search_purchase_history'] = customer_order_df.text_embedding.apply(lambda x: cosine_similarity(x, embeddings_customer_question))

# Sort the DataFrame in descending order based on similarity score
customer_order_df = customer_order_df.sort_values('search_purchase_history', ascending=False)
```

Recommendation Chatbot



- After finding the best recommendation, the result is presented to the users via ChatGPT

Summary

We have covered:

- **NLP-based Applications**
 - Fraud Detection Model
 - Human Trafficking Model
- **ChatGPT-based Applications**
 - Content Moderation
 - Product Recommendation Chatbot

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